

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0804

Homosassa 100 kV
August 01, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0804. The Interconnection Customer proposes a 167.7 MW solar photovoltaic facility interconnecting at the Homosassa 100 kV substation in the Ohio area. The request was received on May 30, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 233 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,380,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
TREGO – HIGHGATE 230 kV	TREGO – TROY ENERGY LLC 500 kV	143.5	395	18.9
TREGO 230/1kV Transformer	TROY ENERGY LLC SUBSTATION – TREGO 500 kV line	111.9	644	21.0
GALEVILLE – GRUVER 230 kV Line	IMPERIAL – EUFAULA 500 kV	103.1	514	7.2
CORRIDOR – LOOKINGGLASS 230 kV	TREGO – BEAVER SUBSTATION 500 kV	100.8	800	10.4
BEAVER 500/1 kV TX	BEAVER – EUFAULA 500 kV	100.6	2,321	12.9
BEAVER 230/1kV Transformer	BEAVER – EUFAULA 500 kV	100.6	2,321	12.9
JOSO – ONANCOCK DIESEL GENERATION FACILITY 230 kV	BEAVER – EUFAULA SUBSTATION 500 kV	100.5	1,738	15.8
CANAL ROAD – NEWPORT TIE SUBSTATION 230 kV	TREGO – BEAVER 500 kV	100.0	1,013	10.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HIGHGATE – TREGO 230 kV	Rebuild line with higher-capacity conductor	\$50,044,000	17% (\$8,642,000)
TREGO 230-1 kV TX	Replace transformer with higher-rated unit	\$6,840,000	10% (\$716,000)
GRUVER – GALEVILLE 230 kV	Reconductor line	\$4,921,000	7% (\$334,000)
CORRIDOR – LOOKINGGLASS 230 kV	Reconductor line	\$5,914,000	5% (\$309,000)
BEAVER 500/1 kV TX	Replace transformer with higher-rated unit	\$46,227,000	5% (\$2,132,000)
BEAVER 230/1kV XFMR	Replace transformer with higher-rated unit	\$10,750,000	5% (\$496,000)
JOSO – ONANCOCK DIESEL GENERATION FACILITY 230 kV	Reconductor line	\$8,921,000	6% (\$497,000)
CANAL ROAD – NEWPORT TIE SUBSTATION 230 kV	Reconductor line	\$5,498,000	5% (\$254,000)
	Total allocated to project		\$13,380,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.